KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5332 | 5 Credits: 4 | Program core | Source-grounded

AI for Robotics

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5332 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5332.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	5332
Course title	AI for Robotics
Semester	5 Credits: 4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

11 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	14 principles and techniques Understand learning concepts and learning from Applying	See official syllabus
CO2	15 outcomes Understand the importance, applicability, and Applying	See official syllabus
CO3	15 strength of AI. Familiarize with programming languages in AI and Understanding	See official syllabus
CO4	14 understand the architecture of Expert system.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2 1
CO2	CO2 2 1
CO3	CO3 2 1
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Familiarize students with Artificial Intelligence principles and techniques	3	As specified
Module 2	Understand learning concepts and learning from outcomes	3	As specified
Module 3	Understand the importance, applicability, and strength of AI.	3	As specified
Module 4	architecture of Expert system.	2	As specified

MODULE 1

Familiarize students with Artificial Intelligence principles and techniques

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize students with Artificial Intelligence principles and techniques
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

4 Understanding historical background

4 Understanding historical background is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding historical background using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding historical background should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding historical background means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding historical background □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Understand various application areas of AI

Understand various application areas of AI is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand various application areas of AI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand various application areas of ai should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand various application areas of AI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understand various application areas of AI □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview

Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand about the basics of expert systems 4 understanding artificial intelligence - introduction, its importance, the turing test, foundations of artificial intelligence, a brief historical overview. application areas of ai - vision and speech processing, robotics, expert systems -basic overview should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Familiarize students with Artificial Intelligence principles and techniques

            Familiarize with basic AI concepts and
  M1.01                                                           4
Understanding
            historical background
  M1.02     Understand various application areas of AI            6
Understanding

  M1.03     Understand about the basics of expert systems         4
Understanding

Artificial intelligence - Introduction, its importance, The Turing test,
Foundations of
artificial intelligence, A brief historical overview. Application areas of AI -
vision and
speech processing, robotics, expert systems -basic overview
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Understand learning concepts and learning from outcomes

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand learning concepts and learning from outcomes
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Understand the basic types of learning

Understand the basic types of learning is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the basic types of learning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the basic types of learning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the basic types of learning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Understand the basic types of learning [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used

Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the details of reinforcement learning 3 understanding familiarize with programming languages used should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG.

6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG. is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying in ai learning - forms of learning, supervised learning algorithms, unsupervised learning algorithms, reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;programming language: introduction to programming language, lisp and prolog. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand learning concepts and learning from outcomes

  M2.01     Understand the basic types of learning                6
Understanding

  M2.02     Discuss the details of reinforcement learning         3
Understanding

            Familiarize with programming languages used
  M2.03                                                           6
Applying
            in AI

            Series Test – I                                       1

Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning
Algorithms, Reinforcement based learning - overview with basic elements agent,
environment, action, state, reward only;Programming Language: Introduction to
programming Language, LISP and PROLOG.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Understand the importance, applicability, and strength of AI.

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the importance, applicability, and strength of AI.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding expert systems Understand the knowledge representation

5 Understanding expert systems Understand the knowledge representation is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding expert systems Understand the knowledge representation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding expert systems understand the knowledge representation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding expert systems Understand the knowledge representation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 5 Understanding expert systems Understand the knowledge representation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding schemes in expert systems

4 Understanding schemes in expert systems is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding schemes in expert systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding schemes in expert systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding schemes in expert systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Understanding schemes in expert systems [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the

Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize with expert system tools 6 expert systems- definition – features of an expert system – organization – characteristics – prospector – knowledge representation in expert systems – expert system tools – mycin – emycin. familiarize with programming languages in ai and understand the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand the importance, applicability, and strength of AI.

            Discuss different features and organization of
  M3.01                                                           5
Understanding
            expert systems
            Understand the knowledge         representation
  M3.02                                                           4
Understanding
            schemes in expert systems

  M3.03     Familiarize with expert system tools                  6

EXPERT SYSTEMS- Definition – Features of an expert system – Organization –
Characteristics – Prospector – Knowledge Representation in expert systems – Expert
system tools – MYCIN – EMYCIN.
              Familiarize with programming languages in AI and understand the
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

architecture of Expert system.

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: architecture of Expert system.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

10 robotics

10 robotics is an official topic in Module 4 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 10 robotics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 10 robotics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 10 robotics means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 10 robotics [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002

Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002 is an official topic in Module 4 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

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- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the ethics and risks of ai usage in robotics 4 ai in robotics and applications- robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, ethics and risks of artificial intelligence in robotics text / reference t/r book title/author stuart russel, peter norvig, “artificial intelligence: a modern approach” pearson t1 education, india,2016 nilsson, nils j.“principles of artificial intelligence” springer science & business t2 media. deepak khemani, a first course in artificial intelligence, mcgraw hill education (india) r1 2013 n.p.padhy: artificial intelligence and intelligent systems, oxford university press, r2 2009. negnevitsky, m, “artificial intelligence: a guide to intelligence systems”, r3 harlow:addisonwesley,2002 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

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Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, R3 Harlow:AddisonWesley,2002 □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
architecture of Expert system.
              Familiarize with AI applications pertaining to
   M4.01                                                    10
              robotics
   M4.02      Discuss the ethics and risks of AI usage in robotics          4

              Series Test – II                                              1
 AI In Robotics and Applications- Robotic perception, localization, mapping-
configuring
 space, planning uncertain movements, dynamics and control of movement, Ethics
and risks
 of artificial intelligence in robotics

Text / Reference

 T/R                                      Book Title/Author


        Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach”
Pearson
  T1
        Education, India,2016
        Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science
& Business
  T2
        Media.
        Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill
Education (India)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 4 Understanding historical background. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding historical background*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Understand various application areas of AI. (3 marks)

Answer: Begin with the standard definition or identity of *Understand various application areas of AI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview. (3 marks)

Answer: Begin with the standard definition or identity of *Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain Understand the basic types of learning. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the basic types of learning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG.. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only;Programming Language: Introduction to programming Language, LISP and PROLOG.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 5 Understanding expert systems Understand the knowledge representation. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding expert systems Understand the knowledge representation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding schemes in expert systems. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding schemes in expert systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the. (3 marks)

Answer: Begin with the standard definition or identity of *Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 10 robotics. (3 marks)

Answer: Begin with the standard definition or identity of *10 robotics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, R3 Harlow:AddisonWesley,2002. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, R3 Harlow:AddisonWesley,2002*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Understanding historical background**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Understand the basic types of learning**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding expert systems Understand the knowledge representation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **10 robotics**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5332. Verify institutional updates against the current official curriculum.